

AE 23: Multiple Logistic Regression

! Important

- Open [RStudio](#) and create a subfolder in your AE folder called “AE-23”.
- Go to the [Canvas](#) and locate your AE-23 assignment to get started.
- Upload the `ae-23.qmd` and `framingham.csv` files into the folder you just created. The `.qmd` and PDF responses are due in Canvas. You can check the due date on the Canvas assignment.

Packages

```
library(tidyverse)
library(broom)
library(ggformula)
library(mosaic)
library(knitr)
```

Data: Framingham study

This data set is from an ongoing cardiovascular study on residents of the town of Framingham, Massachusetts. We want to predict if a randomly selected adult is high risk for heart disease in the next 10 years.

Response variable

- TenYearCHD:
 - 1: Patient developed heart disease within 10 years of exam
 - 0: Patient did not develop heart disease within 10 years of exam

What are my predictor variables?

Based on your group, use the following as your predictor variables.

- Group 1:
 - BMI: patient's body mass index
 - diabetes: 1 if patient has diabetes, 0 otherwise
- Group 2:
 - cigsPerDay: number of cigarettes patient smokes per day
 - currentSmoker: 1 if current smoker, 0 otherwise
- Group 3:
 - sysBP: systolic blood pressure (mmHg)
 - BPMeds: 1 if they are on blood pressure medication, 0 otherwise
- Group 4:
 - totChol: total cholesterol (mg/dL)
 - prevalentHyp: 1 if patient was hypertensive, 0 otherwise

Exercise 0

Edit the code below to drop any missing values, convert the categorical predictors into factors, and select only the eight variables listed above along with `TenYearCHD`. Make sure to do those thing in an order that makes sense:

```
heart_disease <- read_csv("framingham.csv")
```

Exercise 1

Fit a logistic regression model predicting `TenYearCHD` using your two assigned predictors. Write your resulting model on the board in any format you choose.

Exercise 2

For both of your coefficients complete the madlibs below interpreting the value, you may need to edit a bit to make the final sentence grammatically correct:

Quantitative variable: When [variable name] increases by [number], the typical [increase/decrease] in [quantity] is [number], [qualifying statement that makes this different than when we have one predictor].

Quantitative variable: When [variable name] increases by [number], the typical odds-ratio is multiplied by a factor of [number], [qualifying statement that makes this different than when we have one predictor].

Categorical variable: For patients who are [insert category] at the time of examination, the typical [increase/decrease] in [quantity] is [number] vs patients [insert opposite category], [qualifying statement that makes this different than when we have one predictor].

Categorical variable: For patients who are [insert category] at the time of examination, the odds-ratio typically [increases/decreases] by a factor of [number] vs patients [insert opposite category], [qualifying statement that makes this different than when we have one predictor].

Exercise 3

Compute p-values for both of your coefficients using `tidy`.

Exercise 4

For both of your coefficients write the null and alternative hypotheses (in both symbols and words), the test statistics, and the p-values on the board (just pick them out from the `tidy` output). Then, complete the madlibs below interpreting the p-values, you may need to edit the final statements to make them grammatically correct:

Quantitative variable: The [quantity] is [size], so we [do something to] H_0 . The data [do/do not] provide sufficient evidence that [variable name] is a statistically significant predictor of [variable], [qualifying statement that makes this different than when we have one predictor].

Categorical variable: The [quantity] is [size], so we [do something to] H_0 . The data [do/do not] provide sufficient evidence that [variable name] is a statistically significant predictor of [variable], [qualifying statement that makes this different than when we have one predictor].

Exercise 5

Use `tidy` to compute 99% confidence intervals for both of your coefficients. Write them on the board.

Exercise 6

Interpret the the confidence intervals below:

Log-odds interpretation (Quantitative):

Odds-ratio interpretation (Quantitative):

Log-odds interpretation (Categorical):

Odds-ratio interpretation (Categorical):

Exercise 7

1. Fit the full model. I.e. the model using all eight predictors included in the data set.
2. Use `glance` to compute the deviance of both models. Which has the better deviance?
3. Use the `anova` function to conduct a likelihood ratio test between the model containing your two predictors and the full model.
4. On the white board, write the null and alternative hypotheses (in both symbols and words), the test statistic, the p-value on the board. 5 What is the conclusion of your test in context?

Submission

! Important

To submit the AE:

- Render the document to produce the PDF with all of your work from today's class.
- Upload your QMD and PDF files to the Canvas assignment.